

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)			<u>{Vibrations / oscillations}</u> (1) Don't accept moves are perpendicular to {wave / energy transfer / propagation} direction in transverse waves and parallel to wave direction in longitudinal waves (1)	2			2		
	(b)	(i)		Total internal reflection / TIR	1			1		
		(ii)		Air has a lower density [than glass] or it moves [from a more dense] towards a less dense [material] (1) Accept moving towards a material in which it would travel faster Light hits the side above {the critical angle / 42°} (1)	2			2		
	(c)			S waves travel through solids but not liquids or S waves can only travel through solids (1) So the <u>mantle</u> is solid and the <u>core</u> is liquid (1) Alternative: S waves travel through the solid <u>mantle</u> (1) but not the liquid <u>core</u> (1) Reference to inner and outer treat as neutral	2			2		
				Question 5 total	7	0	0	7	0	0